

Causal inference with longitudinal treatment

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Longitudinal study

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Longitudinal study

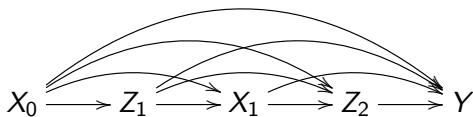
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- A series of work by Robins in biostatistics

Setup

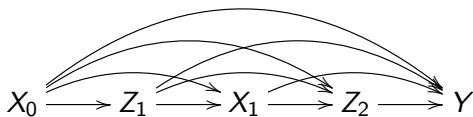
- Temporal order with two time points



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- Z_1 : treatment at time point 1
- X_1 : time varying covariates between the treatments at time points 1 and 2
- Z_2 : treatment at time point 2
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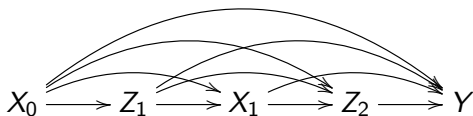
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- Question: what is the causal effect of the treatments at two time points (Z_1, Z_2) on the outcome?
 - Difficulty: X_1 is a post-treatment variable affected by Z_1

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 - allows for latent confounders between X_1 and Y

- Outcome-based identification formula under sequential ignorability

$$\mathbb{E}\{Y_i(z_1, z_2)\} = \mathbb{E}[\mathbb{E}\{\mathbb{E}(Y_i \mid Z_{i1} = z_1, Z_{i2} = z_2, X_{i0}, X_{i1}) \mid Z_{i1} = z_1, X_{i0}\}]$$

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- proof: double expectation conditional on $X_0 \rightarrow$ use ignorability of Z_1
 $\rightarrow$ double expectation conditional on $X_1 \rightarrow$ use ignorability of Z_2

- Integral form ($f(\cdot)$ represents the pdf or pmf of a random variable)

$$\mathbb{E}\{Y_i(z_1, z_2)\} = \iint \mathbb{E}(Y \mid z_1, z_2, x_0, x_1) f(x_1 \mid z_1, x_0) f(x_0) dx_1 dx_0$$

Outcome regression: plug-in estimator

- Plug the estimators of $\mathbb{E}(Y \mid z_1, z_2, x_0, x_1)$, $f(x_1 \mid z_1, x_0)$, and $f(x_0)$ into the identification formula

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- A special case: $\mathbb{E}(Y \mid z_1, z_2, x_0, x_1) = \beta_0 + \beta_1 z_2 + \beta_2 z_1 + \beta_3 x_1 + \beta_4 x_0$

$$\mathbb{E}\{Y_i(z_1, z_2)\} = \beta_0 + \beta_1 z_2 + \beta_2 z_1 + \beta_3 \int \mathbb{E}(X_{i1} \mid z_1, x_0) f(x_0) dx_0 + \beta_4 \mathbb{E}(X_{i0})$$

$$\mathbb{E}\{Y_i(1, 0) - Y_i(0, 0)\} = \beta_2 + \beta_3 \text{ACE}(Z_1 \rightarrow X_1)$$

$$\mathbb{E}\{Y_i(0, 1) - Y_i(0, 0)\} = \beta_1$$

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- Estimate $\text{ACE}(Z_1 \rightarrow X_1)$ as what we did before
 - under linear model $\mathbb{E}(X_{i1} \mid z_1, x_0) = \gamma_0 + \gamma_1 z_1 + \gamma_2 x_0$,
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- g-null paradox: the hypothesis of zero causal effect may be falsely rejected under model misspecification

Recursive estimation

- Relationship between the conditional expectations of $Y(z_1, z_2)$

$$\begin{aligned}\mathbb{E}\{Y_i(z_1, z_2)\} &= \mathbb{E}[\mathbb{E}\{Y_i(z_1, z_2) \mid X_0\}] \\ \mathbb{E}\{Y_i(z_1, z_2) \mid X_0\} &= \mathbb{E}\{Y_i(z_1, z_2) \mid z_1, X_0\} \\ &= \mathbb{E}[\mathbb{E}\{Y_i(z_1, z_2) \mid z_1, X_0, X_1\} \mid z_1, X_0]\end{aligned}$$

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- Recursive estimation
 - estimate $\mathbb{E}\{Y_i(z_1, z_2) \mid z_1, X_0, X_1\}$ for units with $Z_{i1} = z_1$ by fitting a model of Y_i on (X_1, X_0) for units with $(Z_{i1} = z_1, Z_{i2} = z_2)$, denoted by $\hat{\mathbb{E}}\{Y_i(z_1, z_2) \mid z_1, X_0, X_1\}$

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- estimate $\mathbb{E}\{Y_i(z_1, z_2) \mid X_0\}$ for all units by fitting a model of $\hat{\mathbb{E}}\{Y_i(z_1, z_2) \mid z_1, X_0, X_1\}$ on X_0 for units with $Z_{i1} = z_1$, denoted by $\hat{\mathbb{E}}\{Y_i(z_1, z_2) \mid X_0\}$

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- estimator of $\mathbb{E}\{Y_i(z_1, z_2)\}$ is $n^{-1} \sum_{i=1}^n \hat{\mathbb{E}}\{Y_i(z_1, z_2) \mid X_0\}$
- It is often difficult to specify the model of the fitted conditional expectations

Inverse probability weighting

- Identification formula based on treatment probability

$$\mathbb{E}\{Y_i(z_1, z_2)\} = \mathbb{E}\left\{\frac{\mathbf{1}(Z_{i1} = z_1, Z_{i2} = z_2)Y_i}{e(z_1, X_{i0})e(z_2, Z_{i1}, X_{i0}, X_{i1})}\right\}$$

$$e(z_1, X_{i0}) = \Pr(Z_{i1} = z_1 \mid X_{i0})$$

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 - two versions: HT and Hajek
 - unstable if the estimated treatment probabilities close to 0

Structural model

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- Under sequential ignorability, for any function h

$$\mathbb{E}\{h(Y_i(z_1, z_2), X_{i0}, X_{i1}(z_1))\} = \mathbb{E}\left\{\frac{\mathbf{1}(Z_{i1} = z_1, Z_{i2} = z_2)h(Y_i, X_{i0}, X_{i1})}{e(z_1, X_{i0})e(z_2, Z_{i1}, X_{i0}, X_{i1})}\right\}$$

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- Specify a model of $Y_i(z_1, z_2)$ and then obtain estimating equations using inverse probability weighting

Summary

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 - generalizes the results with a single time point
 - various estimation methods

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 - generalizes the results with a single time point
 - various estimation methods
- Extensions
 - multiple time points
 - structural nested model
 - multiply robust estimation
 - other estimation methods: calibration, matching, TMLE
 - with latent confounders: instrumental variable estimation
 - noncompliance, units dropout

Suggested readings

- General introduction
 - Ding's book, Chapter 29
 - Heran and Robins' book, Chapter 19
- Marginal structural model
 - Robins et al. 2000 "Marginal Structural Models and Causal Inference in Epidemiology"
- Structural nested model
 - Vansteelandt and Joffe. 2014 "Structural nested models and G-estimation: the partially realized promise"
 - Naimi et al. 2017 "An introduction to g methods"